

Unit 4: Applications of Data Warehousing and Data Mining in Government (6 Hrs.)

What is Data Warehousing?

Data Warehouse, it can be explained as a subject-oriented, time-variant, non-volatile, an integrated collection of data. The introduction to Data Warehousing also comprises compiled data from external sources. The purpose of designing a Warehouse is to analyze and induce business decisions by reporting data at a different aggregate level.

A data warehouse is a data storage system. It usually entails a variety of data kinds acquired from multiple sources for a variety of objectives. The process of storing this data with discipline so that it may be retrieved later is known as data warehousing.

Subject-Oriented

Organizations can use the Data Warehouse to analyze a specific subject area. Suppose you want to see how well your sales team has performed in the last 5 years – you can query your Warehouse, and it'll tell you all you need to know. In this case, “sales” can be treated as a subject.

Time-Variant

Data Warehouses are responsible for storing historical data for organizations. For example, a transaction system can hold the most recent address of a customer, but a Data Warehouse will hold all the previous addresses too. It continuously keeps adding data from various sources, apart from keeping the historical data – that's what makes it a time-variant model. The data stored will always vary with time.

- **Non-Volatile**

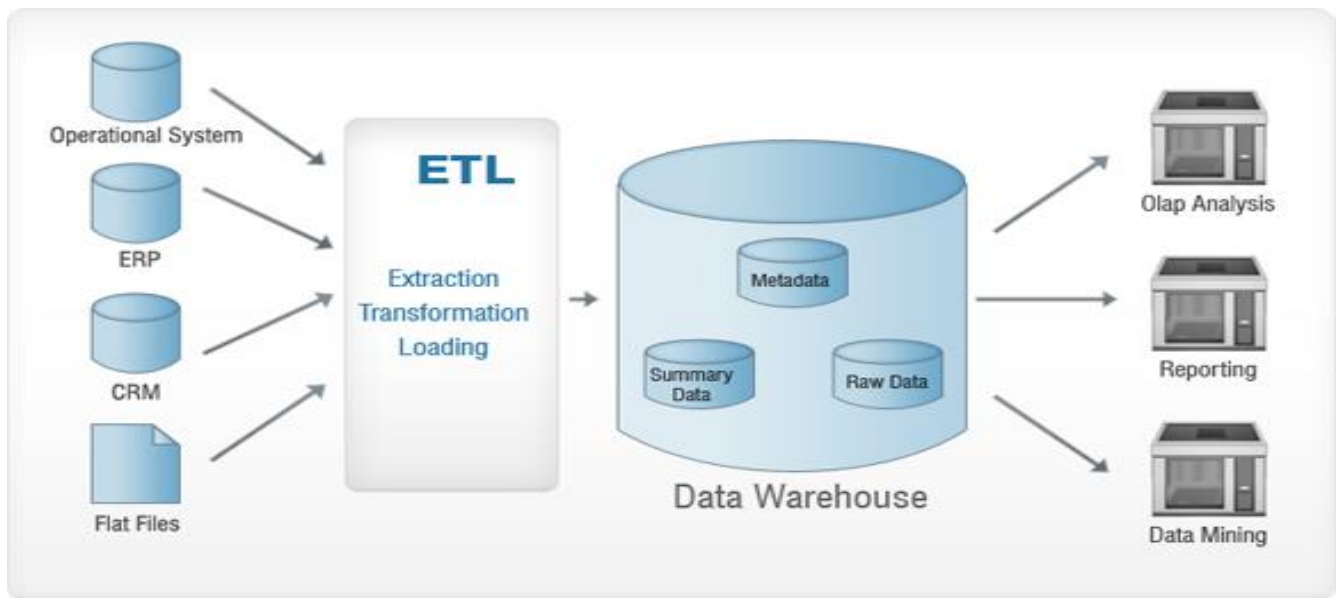
Once data is stored in a Data Warehouse, it can't be altered or modified. We can only add a modified copy of the data we want to modify.

- **Integrated:**

As we said earlier, a Data Warehouse holds data from multiple sources. Say we have two data sources – A and B. Both the sources might have completely different types of data stored in them, but when they are brought to a Warehouse, they're made to undergo preprocessing. That is how a Data Warehouse integrates data from a number of sources.

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Data Warehouse Processes



The data that is collected from various sources (operational system, ERP, CRM, Flat Files, etc.) is made to undergo an ETL process before it's inserted into the data warehouse. This is essentially done to remove anomalies, if any, from the data – so that no harm is caused to the Data Warehouse. ETL stands for – Extraction, Transformation, and Loading. Let's have a look at each of these processes in detail. To understand better, we'll use an analogy – think of a gold rush and read on!

- **Extraction**

Extraction is essentially done to collect all the required data from the source systems using as few resources as possible.

Think of this step like panning the river in search of gold nuggets as big as possible.

- **Transformation**

The main aim is to insert the extracted data into the database in a general format. This is because different sources will have different formats of storing the data – for example, one data source might have data in “dd/mm/yyyy” format, and the other might have it in “dd-mm-yy” format. In this step, we'll convert this into a generalized format – one that'll be used for data from all the sources.

- **Loading**

In this step, the transformed data is loaded into the target database. The process of bringing data from various sources and storing it in the Data Warehouse (after the ETL process, of course), is what is known as Data Warehousing.

Now, you have your data in place – all cleaned up and ready to go. What should be the next step? Extracting knowledge – yes!

Data Mining to the rescue!

[Data Mining to the rescue!](#)

What is Data Mining?

Data Mining is, quite simply, the process of extracting previously unknown but potentially useful information from the data sets. By “previously unknown”, we mean knowledge that can be acquired only after deeply mining the data warehouse – i.e., it won’t make sense on the surface. Data Mining essentially searches for the relationships global patterns that exist between the data elements.

For example, Data Mining helps the supermarket and retail sector owners to know the choices of the customers. Looking at the purchase history of the customers, the data mining tools show the buying preferences of the customers.

With the help of these results, the supermarkets design the placements of products on shelves and bring out offers on items such as coupons on matching products, and special discounts on some products.

Mobile service providers use data mining to design their marketing campaigns and to retain customers from moving to other vendors. From a large amount of data such as billing information, email, text messages, web data transmissions, and customer service, the data mining tools can predict “churn” that tells the customers who are looking to change the vendors. With these results, a probability score is given. The mobile service providers are then able to provide incentives, offers to customers who are at higher risk of churning. This kind of mining is often used by major service providers such as broadband, phone, gas providers, etc.

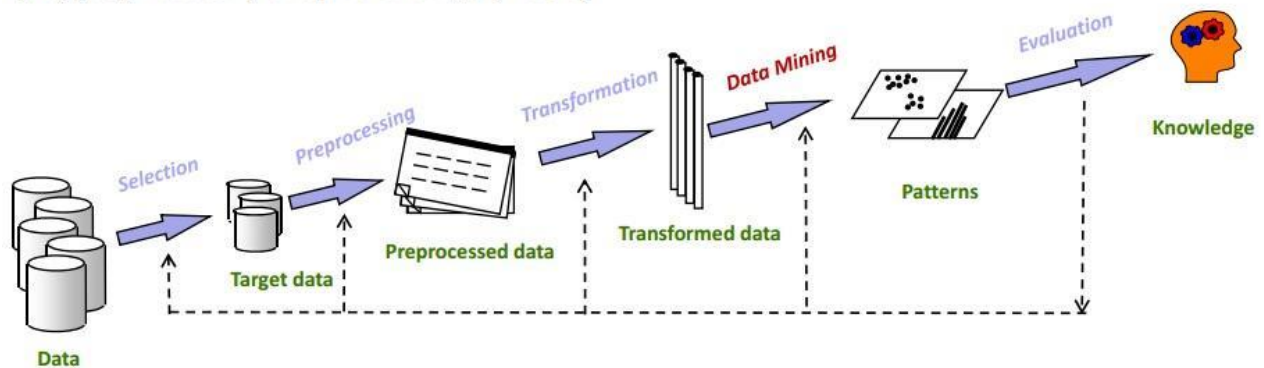
We feel it’ll be unfair to you if we give you the flavor of Data Warehousing and Data Mining and completely ignore the big picture – Knowledge Discovery in Databases (KDD). Data Mining forms one of the steps of a KDD process. Let’s talk a bit more about KDD.

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Knowledge Discovery In Databases (KDD)

Data mining is one of the more crucial steps in the process of KDD. KDD basically covers everything from the selection of data to finally evaluating the mined data. The complete KDD cycle is shown in the image below:

[Fayyad, Piatetsky-Shapiro & Smyth, 1996]



Selection

It is of utmost importance to know the exact target data. Analyzing Data Mining to Data Warehousing subset is a very important step because removing unrelated data elements will reduce the search space during the Data Mining phase.

Pre-processing

In this step, the selected data is freed from any anomalies and outliers. Basically, the data is completely cleaned in this phase. Like, if there are some missing data fields, they're filled with appropriate values. For example, in the table that stores the details of your organization's employees, suppose there's a column for "Middle Name". Chances are, it'll be empty for many employees. In such a scenario, an appropriate value is chosen (N/A, for ex).

Transformation

This phase attempts to reduce the variety of data elements while preserving the quality of the info.

Data mining

This is the main phase of a KDD process. The transformed data is subjected to data-mining methods like grouping, clustering, regression, etc. This is

done iteratively to bring the best results. Different techniques can be used depending on the requirements.

Evaluation

This is the final step. In this, the obtained knowledge is documented and presented for further analysis. Various Data Visualization tools are used in this step to depicting the acquired knowledge in a beautiful and understandable way.

National Data warehouses

A large number of national data warehouses can be identified from the existing data resources within the central government ministries. These potential subject areas on which data warehouse may be developed at present and in future.

Census Data

- Census data can help planners and others understand their communities' social, economic, and demographic conditions.
- Census data is the primary data used by planners to understand the social, economic, and demographic conditions locally and nationally. We sometimes think the census is unique to the United States, but census data is collected in other countries.
- The total process of collecting, compiling, and publishing demographic, economic, and social data about a specific time to all citizens
- in a country or fixed or defined area part of a country. Most countries also include a housing census.
- A census is a procedure of systematically acquiring and recording information about the members of a given population. It is a regularly occurring and official count of a particular population. The term is used mostly in connection with national population and housing censuses; other common censuses include agriculture, business, and traffic censuses.

Possible uses of Census Data (Examples)

- Economic and social planning
- Education planning
- Health care planning
- Infrastructure (roads, railway) development
- Improving housing conditions
- Providing amenities (water, electricity, communication, etc.)
- Assessment/Evaluation of programs
- Emergency planning and disaster response
- Business decisions by the private sector and individuals
- Administrative purposes (elections, creation of new units, etc.)
- Projection of future populations and their social-economic needs (schools, elderly, immunizations, etc..)

Datawarehouse and data mining in Census Data

The registered general and census commissions decennially compile information of all individuals, villages, population groups, etc. This information is wide-ranging such as the individual slip, a compilation of information individual households .

Primary census abstract and village amenities. Subsequently, a data warehouse was also developed for village amenities for Tamil Nadu. This enables multi-dimensional analysis of the village level data in such as education, health, and infrastructure. As the census compilation is performed once in ten years, the data is quasi-static and therefore no refreshing of the warehouse needs to be done on a periodic basis. Only the new data needs to be either appended to the data warehouse or alternatively, a new data warehouse can be built.

Datawarehouse and data mining in Prices of Essential Commodities

The ministry of food and civil supplies, the government of India, compiles daily data for about 300 observation centers in the entire country on the prices of essential commodities such as rice edible oils, etc. The data is compiled at the district level by the respective state government agencies and transmitted online in Delhi for aggregation and storage.

A data warehouse can be built for this data and the OLAP technique can be applied for its analysis. A data mining and forecasting technique can be applied for advanced forecasting of the actual prices of these essential commodities. The forecasting model can be strengthened for more accurate forecasting by taking into account external factors such as rainfall, the growth rate of population, and inflation.

Data warehouse

- Data warehousing is a collection of tools and techniques using which more knowledge can be driven out from a large amount of data. This helps with the decision-making process and improving information resources.
- Datawarehouse is basically a database of unique data structures that allows the relatively quick and easy performance of complex queries over a large amount of data. It is created from multiple heterogeneous sources.
- A data warehouse is built to store a huge amount of historical data and empowers fast requests over all the data, typically using Online Analytical Processing (OLAP).

Datamining

- Data mining refers to extracting knowledge from large amounts of data. The data sources can include databases, data warehouse, web, etc.
- Data mining refers to the analysis of data. It is the computer-supported process of analyzing huge sets of data that have either been compiled by computer systems or have been downloaded into the computer.
- In the data mining process, the computer analyzes the data and extracts useful information from it. It looks for hidden patterns within the data set and tries to predict future behavior. Data mining is

primarily used to discover and indicate relationships among the data sets.

- Data mining aims to enable business organizations to view business behaviors, trends relationships that allow the business to make data-driven decisions.

The key areas that demand the use of data warehousing and data mining

Agriculture

The agricultural census performed by the Ministry of Agriculture, Government, compiles a large number of agricultural parameters at the national level. District-wise agricultural production area and yield of crops are compiled, analysis, mining, and forecast statistics on the consumption of fertilizers can be turned into a data merge. Data in agricultural inputs such as seeds and fertilizers can also be effectively analyzed in a warehouse. Data from livestock census can be turned into a data warehouse. Land use part statistics can also be analyzed in a warehousing environment. Thus, there is the substantial application of data warehouse housing and data mining techniques in the agricultural sector.

Rural Development

Data on individuals below the poverty line can be built into a data warehouse. Drinking water census data (from the drinking water missions) can be effectively utilized by OLAP and data mining technologies. Monitoring and analysis of progress made on the implementation of rural development programs can also be made using OLAP and data mining technologies.

Health

Community needs assessment data, immunization data, data from national programs on controlling blindness, leprosy, malaria can all used for data warehousing implementation, OLAP, and data mining applications. Generate patient, employee, and financial records share data with other entities, like insurance companies, NGOs, and medical aid services Use data

mining to identify patient trends Provide feedback to physicians on procedures and tests.

Planning

At the planning commission, data warehouses can be built for the state plan data on all sectors, labor, energy, education, trade and industry, five-year plan, etc.

Commerce and Trade

Datalink on trade can be analyzed and converted into a data warehouse. World price monitoring systems can be made to perform better by using data warehousing and data mining technologies.

Education

educational survey data converted into a data warehouse. Various types of analytical queries and reports can be answered. Store and analyze information about faculty and students maintain student portals to facilitate student activities extract information for research grants and assess student demographics integrate information from different sources into a single repository for analysis and strategic decision-making.

Other Sectors

In addition to the above mentioned important applications, there exist a number of other potential application areas for data warehousing and data mining, as follows:

Tourism. Tourist arrival behaviour and preferences; tourism products data; foreign exchange earnings data; and Hotels, Travel and Transportation data.

